

CHULMAN NERYUNGRI, Russian Federation

WMO: 303930

Lat: 56.914N

Lon: 124.912E

Elev: 2816

StdP: 13.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.0	-40.4	-47.5	0.4	-43.8	-43.9	0.5	-40.0	17.4	-18.4	15.8	-19.6	1.3	70	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	80.4	61.4	77.0	60.6	73.6	59.0	64.4	74.1	62.8	72.2	61.2	70.0	7.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	90.6	66.8	59.5	84.4	65.5	57.6	78.7	64.4	31.2	74.1	29.9	72.2	28.7	69.9	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.1	13.6	-47.9	85.3	4.7	3.9	-51.3	88.1	-54.1	90.4	-56.7	92.6	-60.2	95.5		
(5)			-47.9	67.9	4.6	2.1	-51.2	69.4	-53.8	70.6	-56.4	71.8	-59.7	73.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	-22.5	-12.5	4.2	23.5	40.8	57.3	61.8	56.3	40.7	19.6	-6.0	-22.2		
	DBStd	31.74	11.21	10.40	11.16	8.84	8.13	7.78	6.37	7.38	8.23	10.44	14.15	11.50		
	HDD50	11733	2248	1749	1421	796	301	27	4	28	296	943	1681	2239		
	HDD65	16398	2713	2169	1886	1246	749	247	135	280	730	1408	2131	2704		
	CDD50	871	0	0	0	0	17	246	371	222	15	0	0	0		
	CDD65	61	0	0	0	0	0	15	37	9	0	0	0	0		
	CDH74	657	0	0	0	0	5	192	342	117	1	0	0	0		
	CDH80	108	0	0	0	0	0	30	66	12	0	0	0	0		
	WSAvg	6.3	4.2	5.4	6.8	7.7	7.9	6.9	6.3	6.4	6.8	7.1	5.6	4.2		
	PrecAvg	20.9	0.6	0.4	0.6	1.1	1.9	3.4	4.1	3.6	2.4	1.4	0.9	0.7		
(15) Precipitation	PrecMax	25.9	1.2	0.8	1.8	2.3	4.4	7.4	9.4	7.4	5.7	2.7	2.7	1.2		
	PrecMin	14.3	0.2	0.2	0.2	0.4	0.4	0.5	1.1	0.9	0.7	0.6	0.3	0.3		
	PrecStd	2.9	0.2	0.2	0.4	0.5	1.0	1.4	1.6	1.5	0.9	0.6	0.4	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	17.7	34.9	52.1	72.1	83.8	86.1	81.4	69.6	48.2	31.9	10.7		
		MCWB	10.3	15.4	28.6	39.4	52.7	61.3	63.8	62.2	54.3	39.7	27.9	10.2		
	2%	DB	4.0	12.5	29.3	45.2	64.7	79.1	81.2	77.3	62.8	41.7	26.0	4.9		
		MCWB	3.5	11.1	24.8	35.8	48.9	59.8	62.4	61.1	51.2	35.4	23.6	4.4		
	5%	DB	-2.0	7.8	25.0	40.9	59.8	75.4	77.7	73.4	58.3	36.9	20.4	-0.4		
		MCWB	-2.5	6.7	21.2	33.1	46.2	57.5	62.0	59.8	49.4	32.6	18.8	-1.0		
	10%	DB	-6.2	3.0	21.2	37.3	55.2	71.8	74.3	69.5	53.8	33.3	14.8	-5.4		
		MCWB	-6.5	2.2	18.5	30.9	43.8	56.0	60.8	58.1	47.1	30.2	13.7	-5.6		
	0.4%	WB	10.6	15.3	29.2	41.0	53.9	64.5	67.3	65.6	56.7	40.8	28.4	10.0		
		MCDB	10.8	17.3	34.1	49.5	69.1	74.6	76.6	74.5	64.5	47.0	30.6	10.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	3.5	11.1	24.8	36.5	50.1	61.7	65.3	63.6	53.6	36.1	23.6	4.3		
		MCDB	3.9	12.5	29.2	43.8	61.8	74.1	75.0	72.4	59.1	40.8	25.8	4.6		
	5%	WB	-2.4	6.6	21.6	33.7	47.4	59.7	63.6	61.8	50.6	32.9	18.7	-1.1		
		MCDB	-2.1	7.7	24.7	39.5	57.4	71.8	73.2	69.9	56.4	35.9	20.4	-0.7		
	10%	WB	-6.6	2.2	18.4	31.2	44.9	57.9	62.2	59.8	47.6	30.4	13.5	-5.6		
		MCDB	-6.3	2.9	21.1	36.5	53.3	68.6	71.1	67.4	53.0	33.0	14.6	-5.3		
	5% DB	MDBR	11.9	14.9	17.9	17.1	17.1	20.2	18.5	18.6	15.8	13.1	12.4	11.3		
		MCDBR	14.2	17.4	19.6	20.2	25.8	26.1	24.5	25.3	24.7	16.9	14.1	14.2		
		MCWBR	13.9	16.0	15.8	13.0	13.7	12.0	10.6	11.9	14.7	11.6	12.3	14.0		
		MCWBR	14.1	17.2	19.1	18.4	22.9	22.0	19.9	20.6	21.0	14.7	14.1	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.232	0.263	0.294	0.381	0.442	0.415	0.438	0.385	0.327	0.289	0.241	0.221		
		taud	2.069	2.085	2.065	1.897	1.991	2.244	2.192	2.339	2.442	2.266	2.074	1.999		
	Ebn at Noon		207	249	271	259	251	262	252	260	262	238	197	171		
		Edh at Noon	18	28	37	52	51	40	42	34	26	23	17	14		
(44) All-Sky Solar Radiation	RadAvg		267	607	1180	1746	1681	1707	1539	1247	853	521	326	182		
	RadStd		10	28	45	80	156	131	160	128	77	50	21	6		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	+2.23	N/A	N/A	N/A	N/A	-297	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page